

mgcvUI GAM Report

2026-03-28

Model Overview

Metric	Value
R-squared	0.8591
CV R-squared	0.8364
Deviance Explained	86.3%
AIC	-3464.9
BIC	-3214.5
Observations	1488
Smooth Terms	9
Method	REML
Family	gaussian
Fitting Time	1.69 sec

Equation

$$\log_{10}(\widehat{\text{sale_price}}) = -31.2954 + f_1(\text{sale_age}) + f_2(\text{living_sqft}) + f_3(\text{beds_total}) + f_4(\text{baths_total}) + f_5(\text{age}) + f_6(\text{garage_spaces})$$

Family: gaussian (link = identity) **Method:** REML

Smooth function definitions:

Function	Term	Description
f_1	s(sale_age)	cubic regression spline, k = 6
f_2	s(living_sqft)	thin plate regression spline, k = 10
f_3	s(beds_total)	thin plate regression spline, k = 8
f_4	s(baths_total)	cubic regression spline, k = 4
f_5	s(age)	thin plate regression spline, k = 10
f_6	s(garage_spaces)	thin plate regression spline, k = 5
f_7	ti(living_sqft,sale_age)	tensor product smooth, k = 5 x 5
f_8	ti(beds_total,living_sqft)	tensor product smooth, k = 5 x 5
f_9	ti(baths_total,living_sqft)	tensor product smooth, k = 5 x 5

Formula

```
sale_price ~ s(sale_age, bs = "cr", k = 6) + s(living_sqft, bs = "tp") + s(beds_total,
  bs = "tp", k = 8) + s(baths_total, bs = "cr", k = 4) + lot_size + area_id + s(age,
  bs = "tp") + latitude + longitude + s(garage_spaces, bs = "tp", k = 5) + ti(living_sqft,
  sale_age, bs = c("tp", "tp")) + ti(beds_total, living_sqft, bs = c("tp", "tp")) +
  ti(baths_total, living_sqft, bs = c("tp", "tp"))
```

Summary

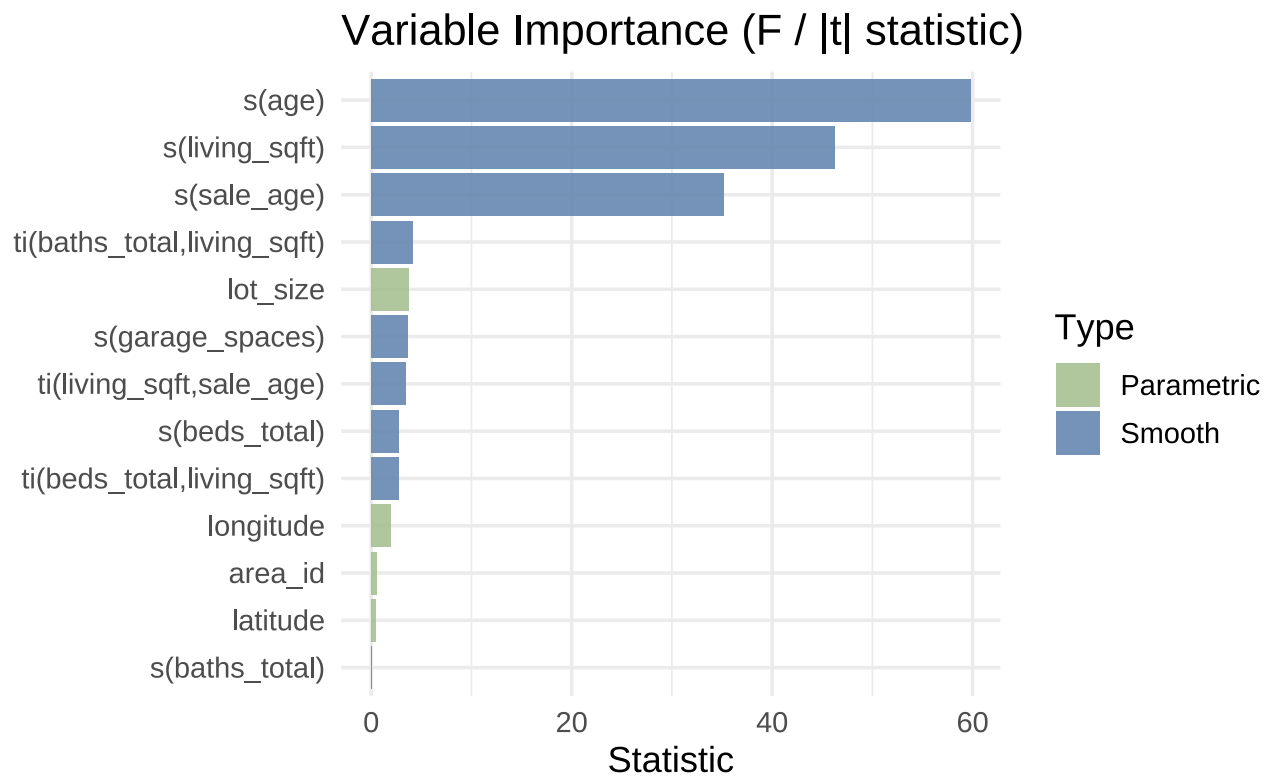
Smooth Terms

Term	EDF	Ref.df	F	p_value
s(sale_age)	4.52	5	35.22	< 2e-16
s(living_sqft)	4.90	9	46.20	< 2e-16
s(beds_total)	2.07	7	2.75	4.95e-07
s(baths_total)	0.00	3	0.00	0.918371
s(age)	7.30	9	59.78	< 2e-16
s(garage_spaces)	2.42	4	3.59	0.000515
ti(living_sqft,sale_age)	4.76	16	3.43	< 2e-16
ti(beds_total,living_sqft)	5.29	16	2.70	< 2e-16
ti(baths_total,living_sqft)	3.03	16	4.15	< 2e-16

Parametric Terms

Term	Estimate	Std_Error	t_value	p_value
(Intercept)	-31.2954	17.8843	-1.750	0.080350
lot_size	0.0000	0.0000	3.746	0.000187
area_id	-0.0005	0.0009	-0.577	0.563766
latitude	-0.1711	0.3440	-0.497	0.618932
longitude	-0.3652	0.1899	-1.923	0.054616

Variable Importance

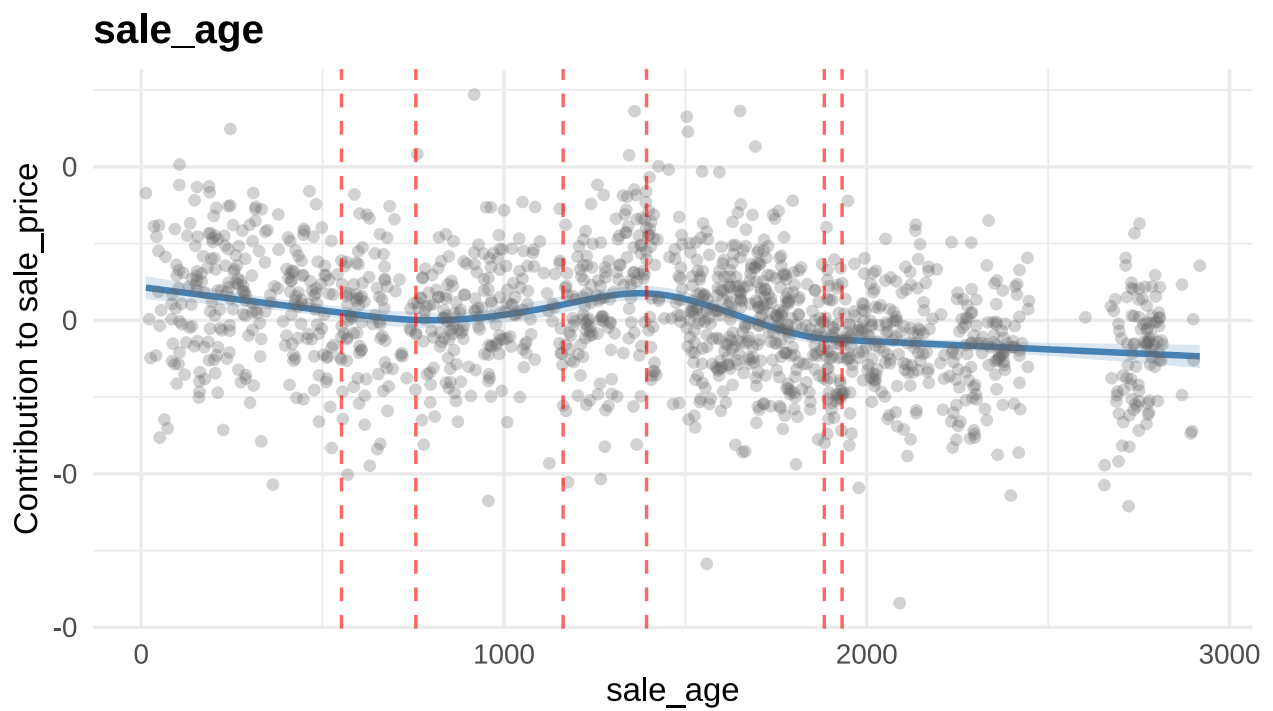


Term	Statistic	Type
s(age)	59.7832	Smooth
s(living_sqft)	46.2018	Smooth
s(sale_age)	35.2169	Smooth
ti(baths_total,living_sqft)	4.1532	Smooth
lot_size	3.7462	Parametric
s(garage_spaces)	3.5927	Smooth
ti(living_sqft,sale_age)	3.4256	Smooth
s(beds_total)	2.7456	Smooth
ti(beds_total,living_sqft)	2.7023	Smooth
longitude	1.9235	Parametric
area_id	0.5774	Parametric
latitude	0.4975	Parametric
s(baths_total)	0.0000	Smooth

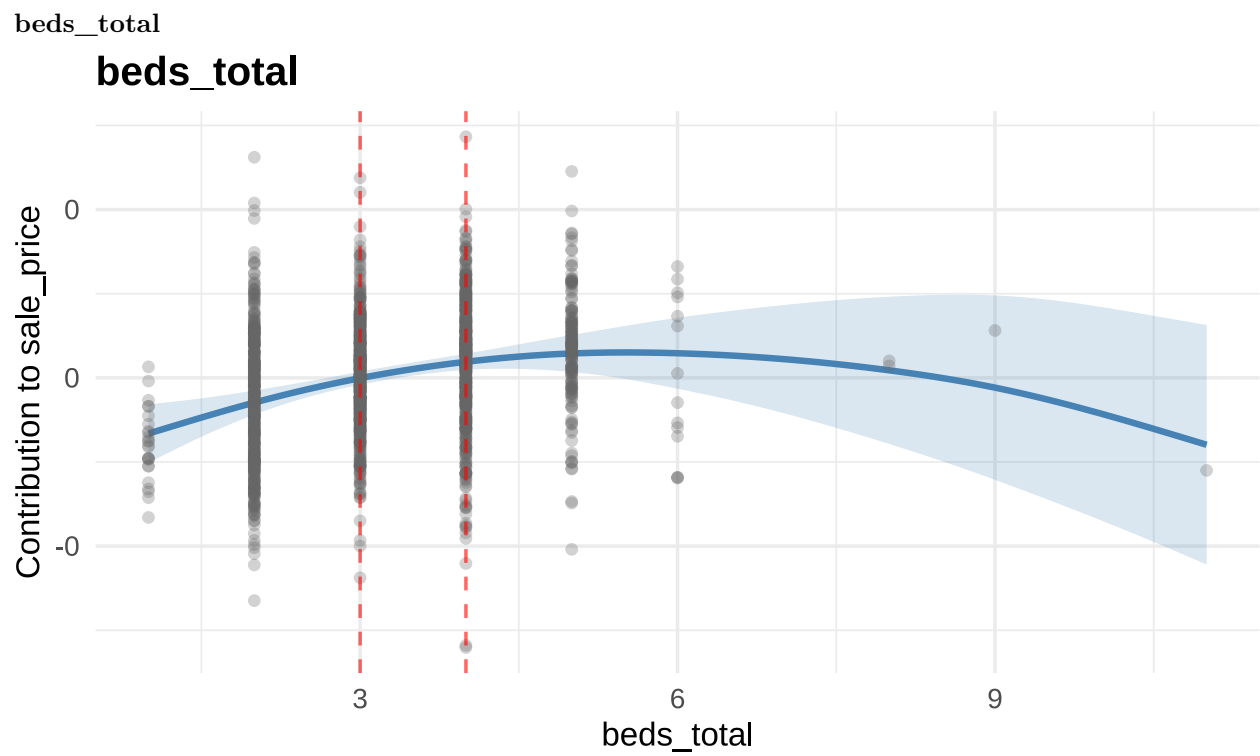
Contribution Plots

Smooth Terms

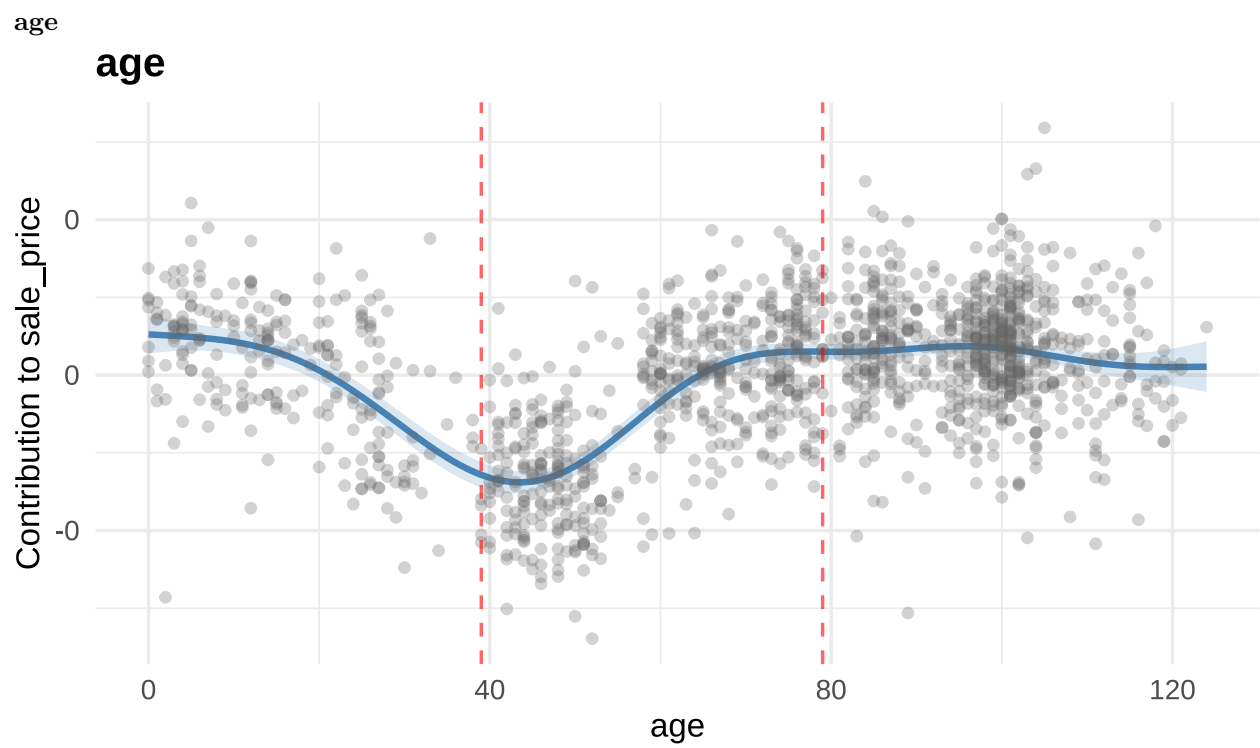
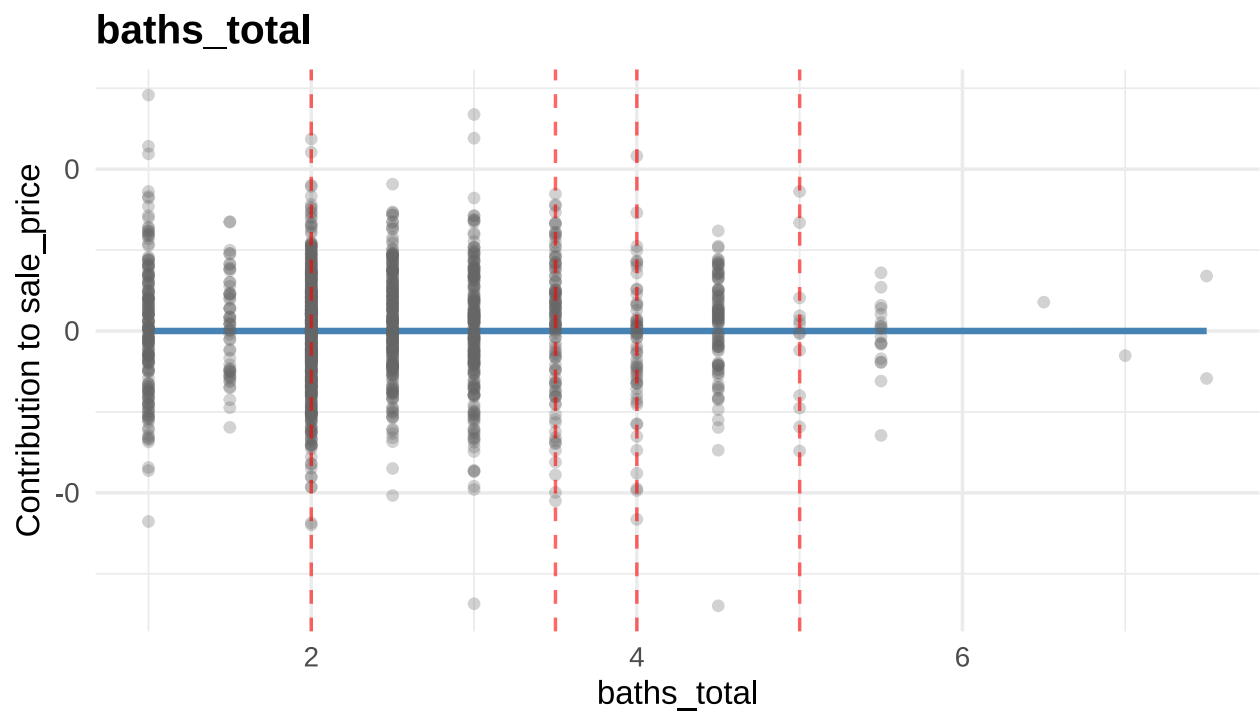
sale_age



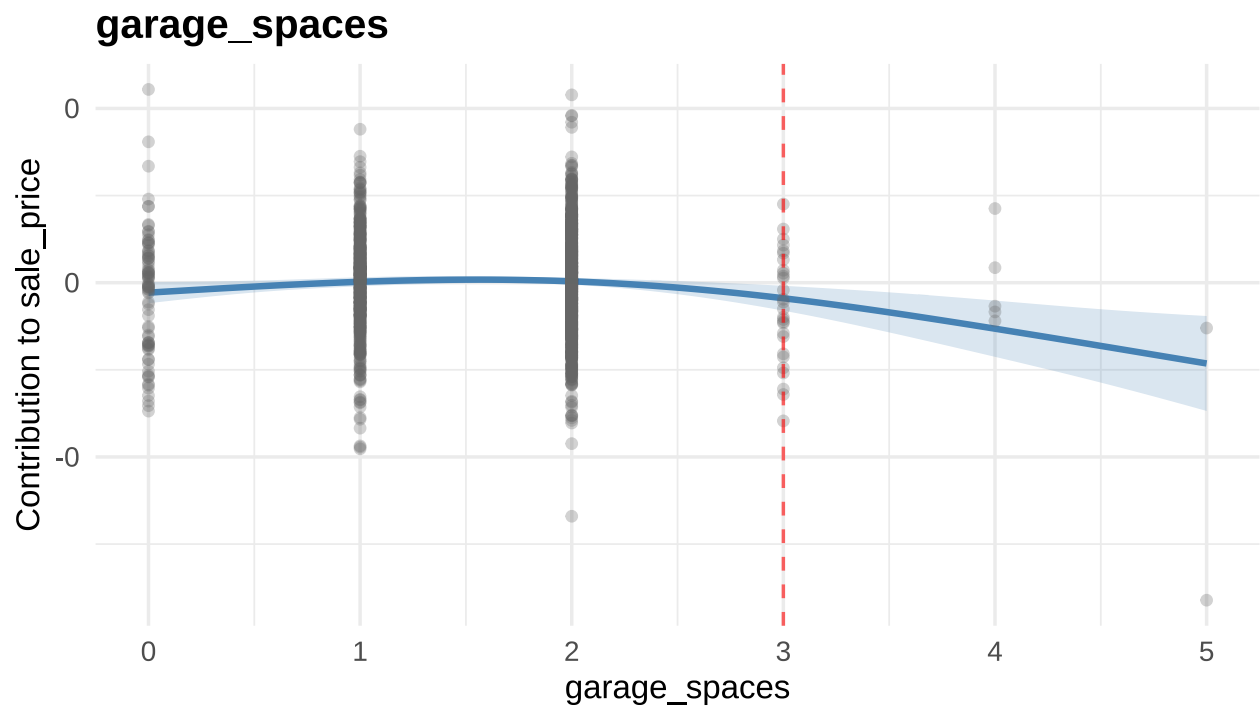
living_sqft



baths_total



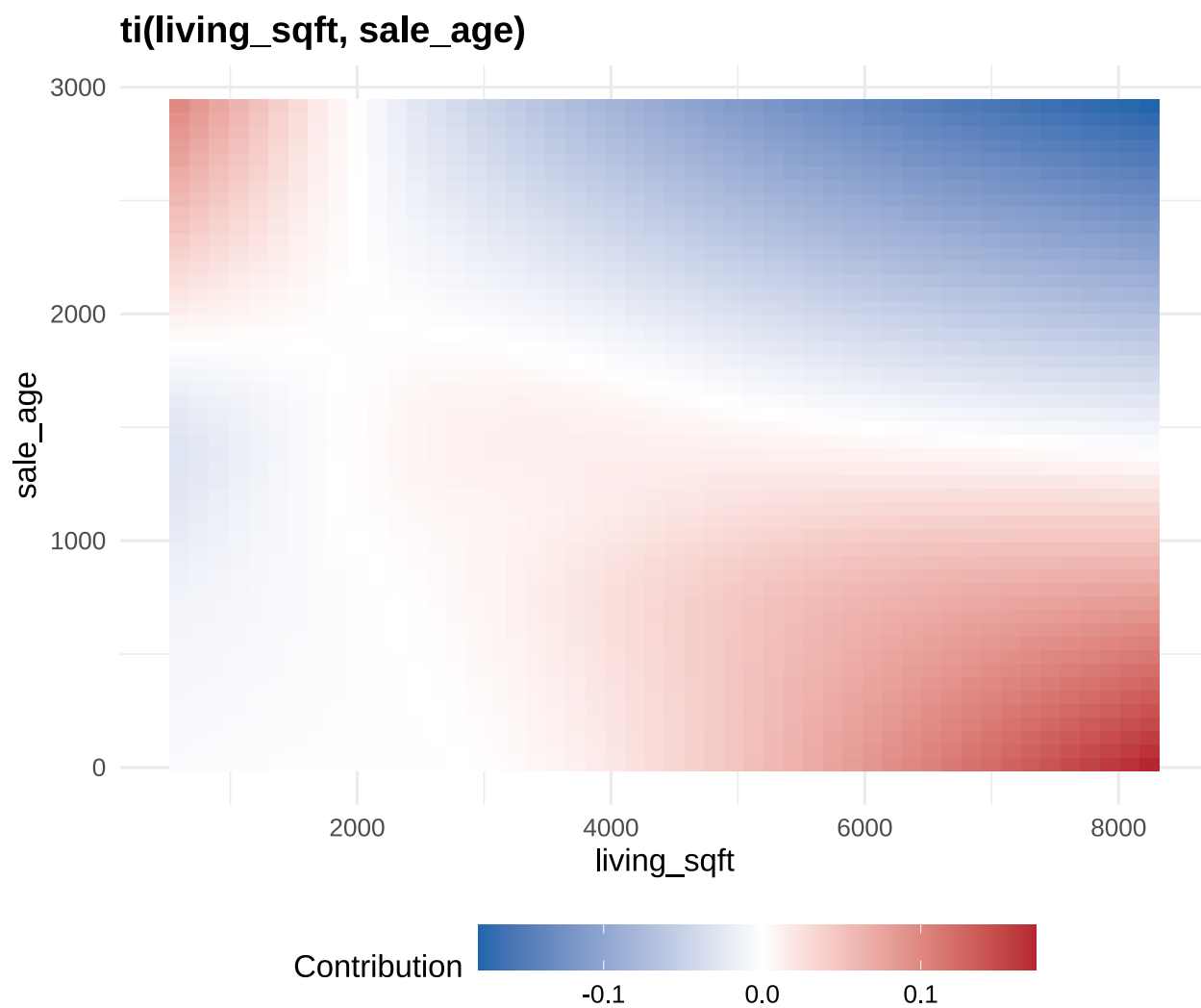
garage_spaces



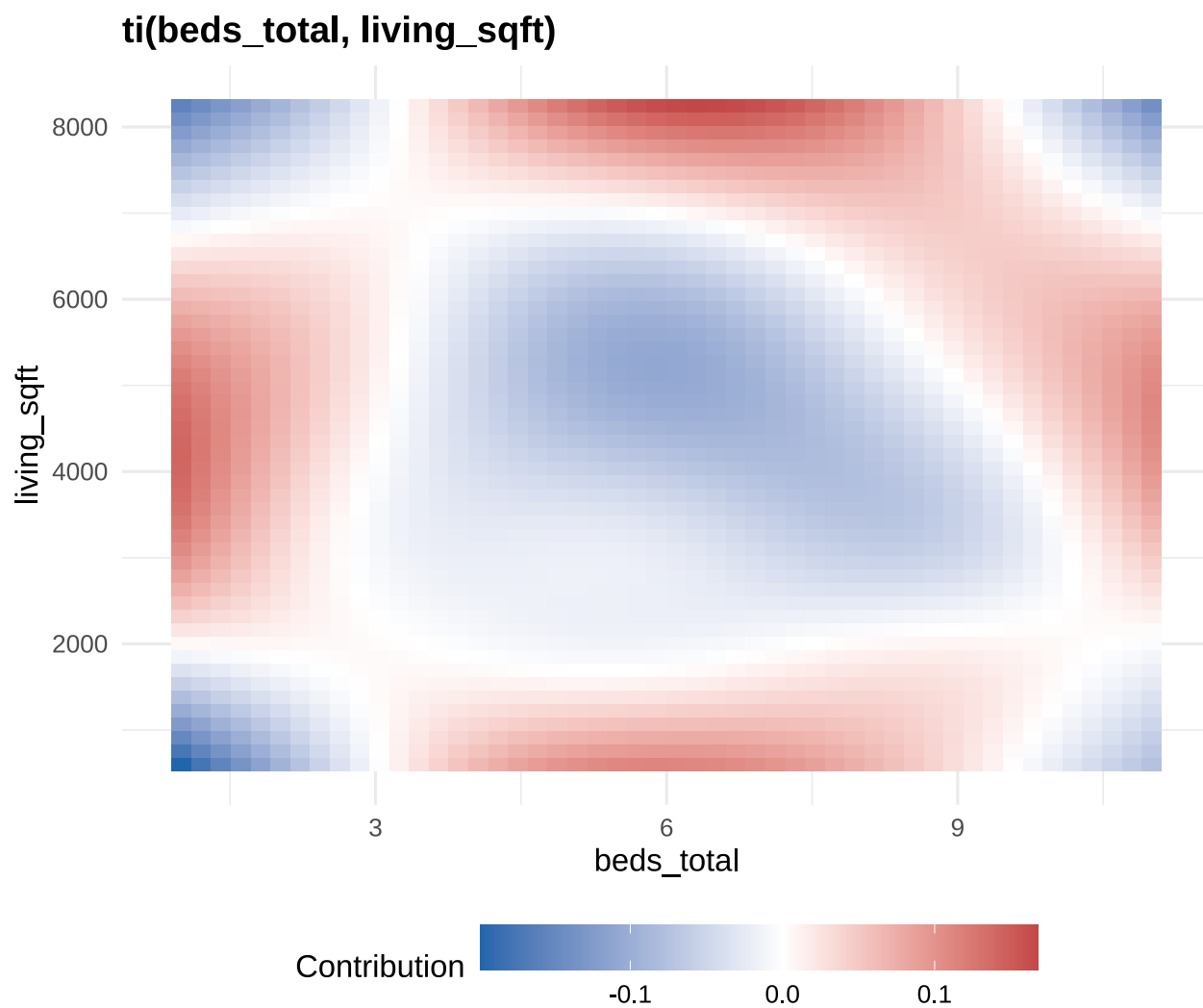
Factor-by-Smooth Terms

Interaction Terms

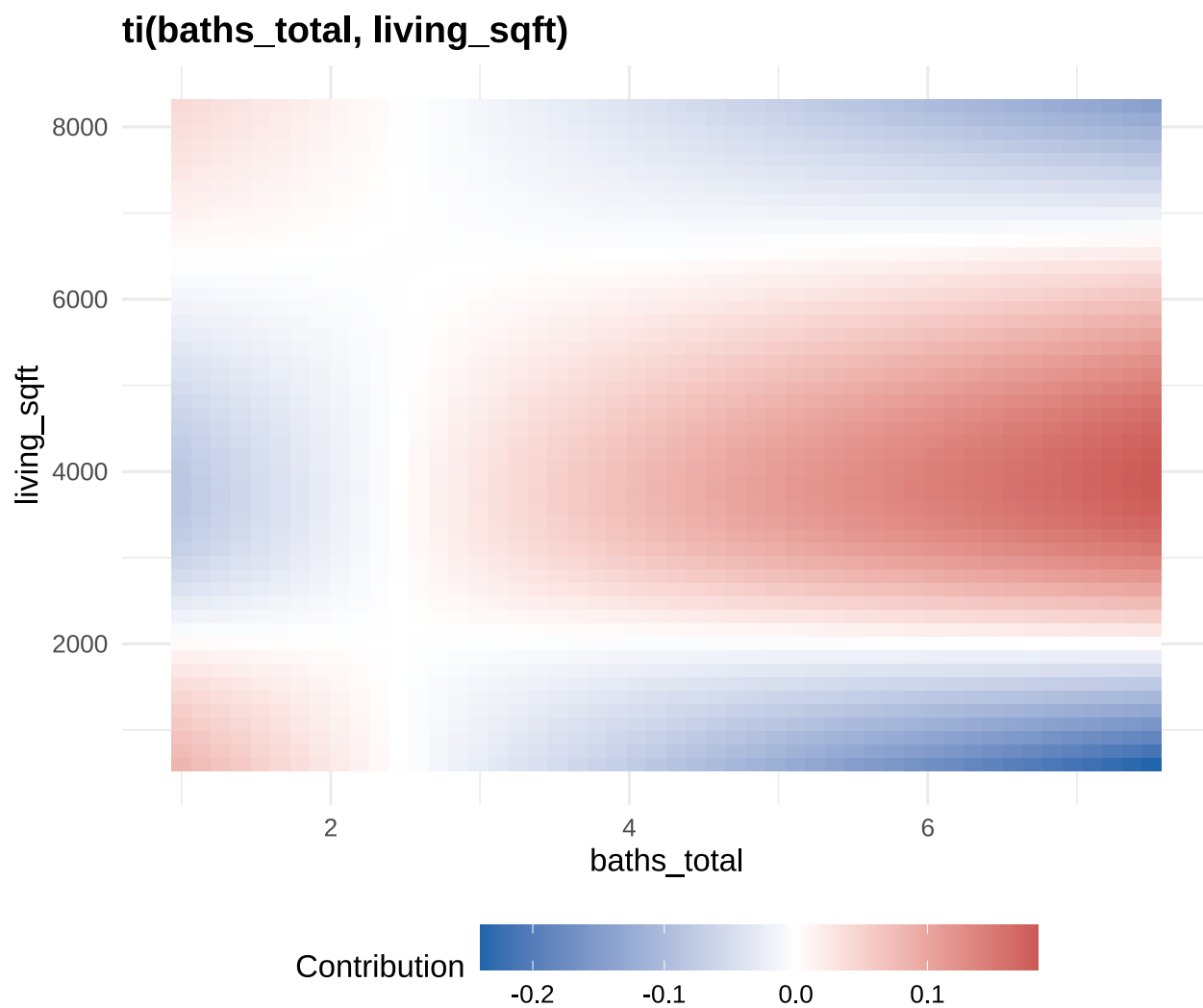
ti(living_sqft, sale_age)



ti(beds_total, living_sqft)



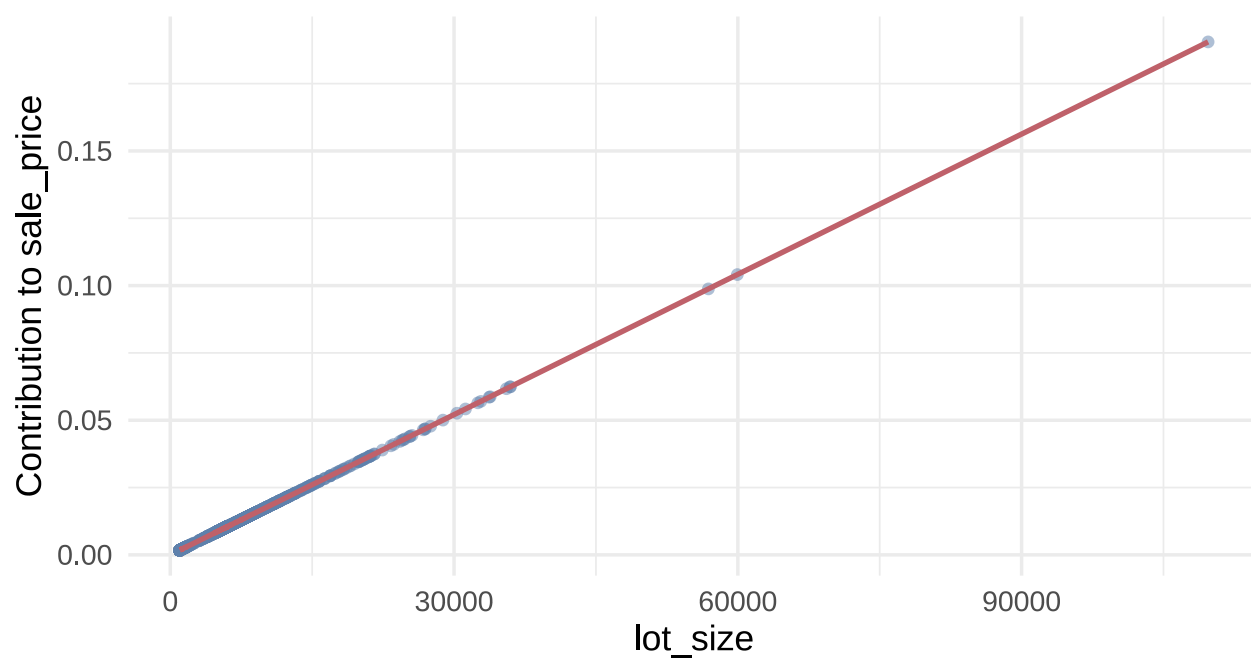
ti(baths_total, living_sqft)



Parametric Terms

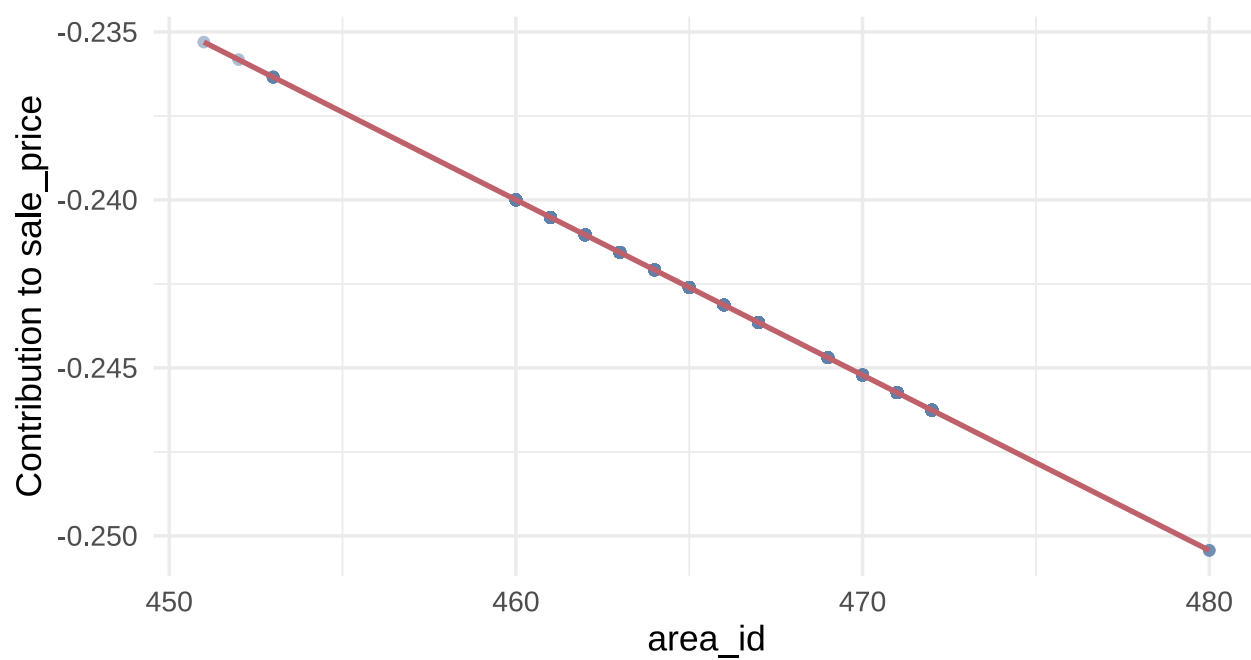
lot_size

lot_size (linear)

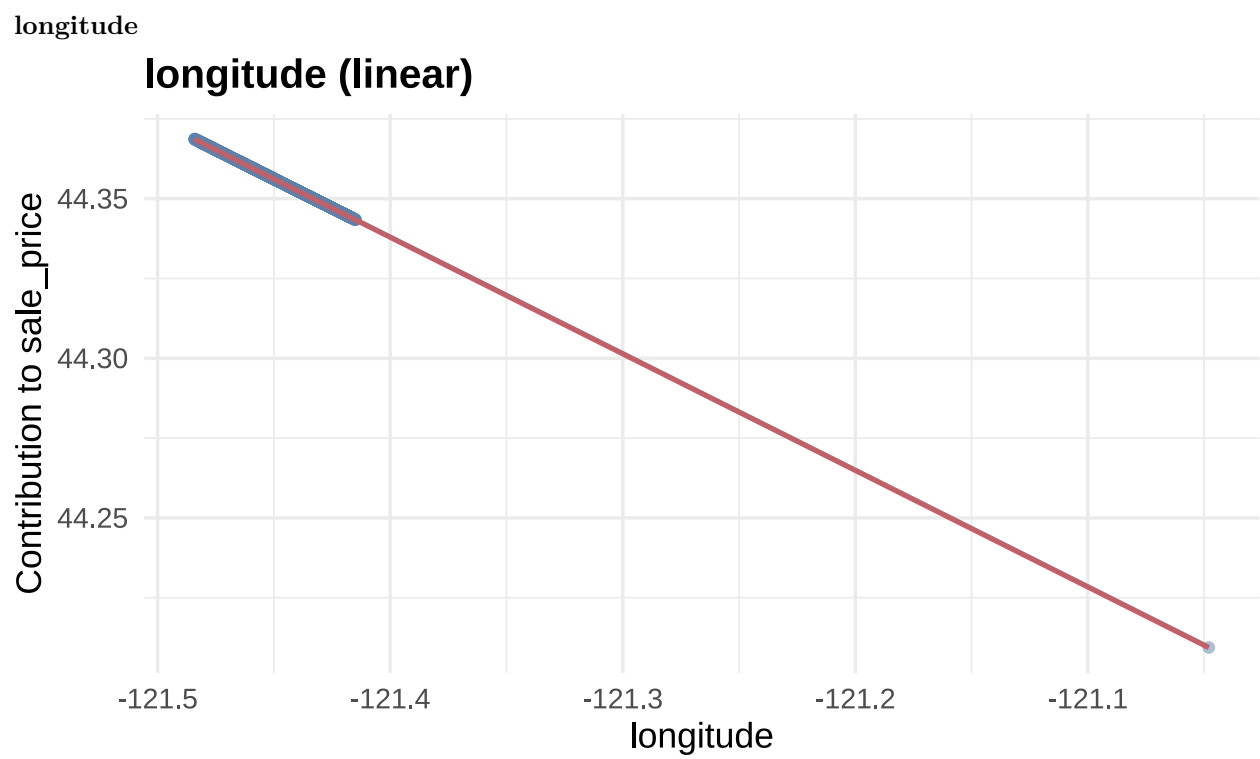
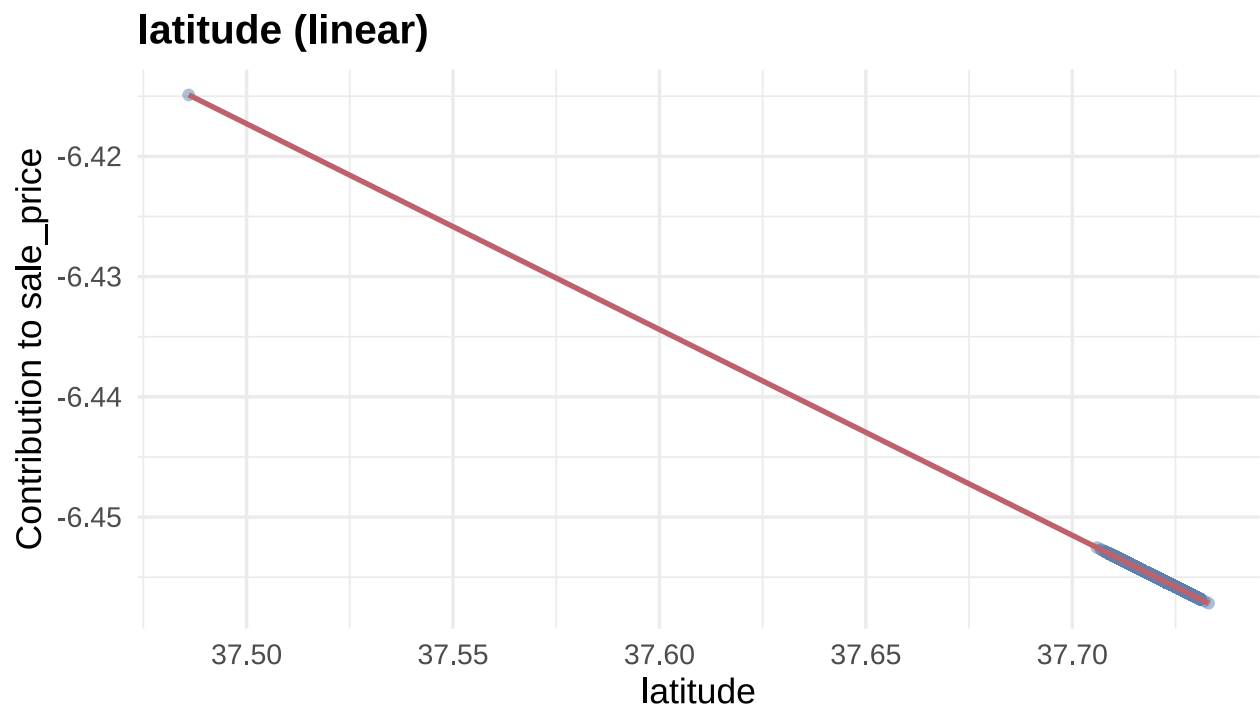


area_id

area_id (linear)

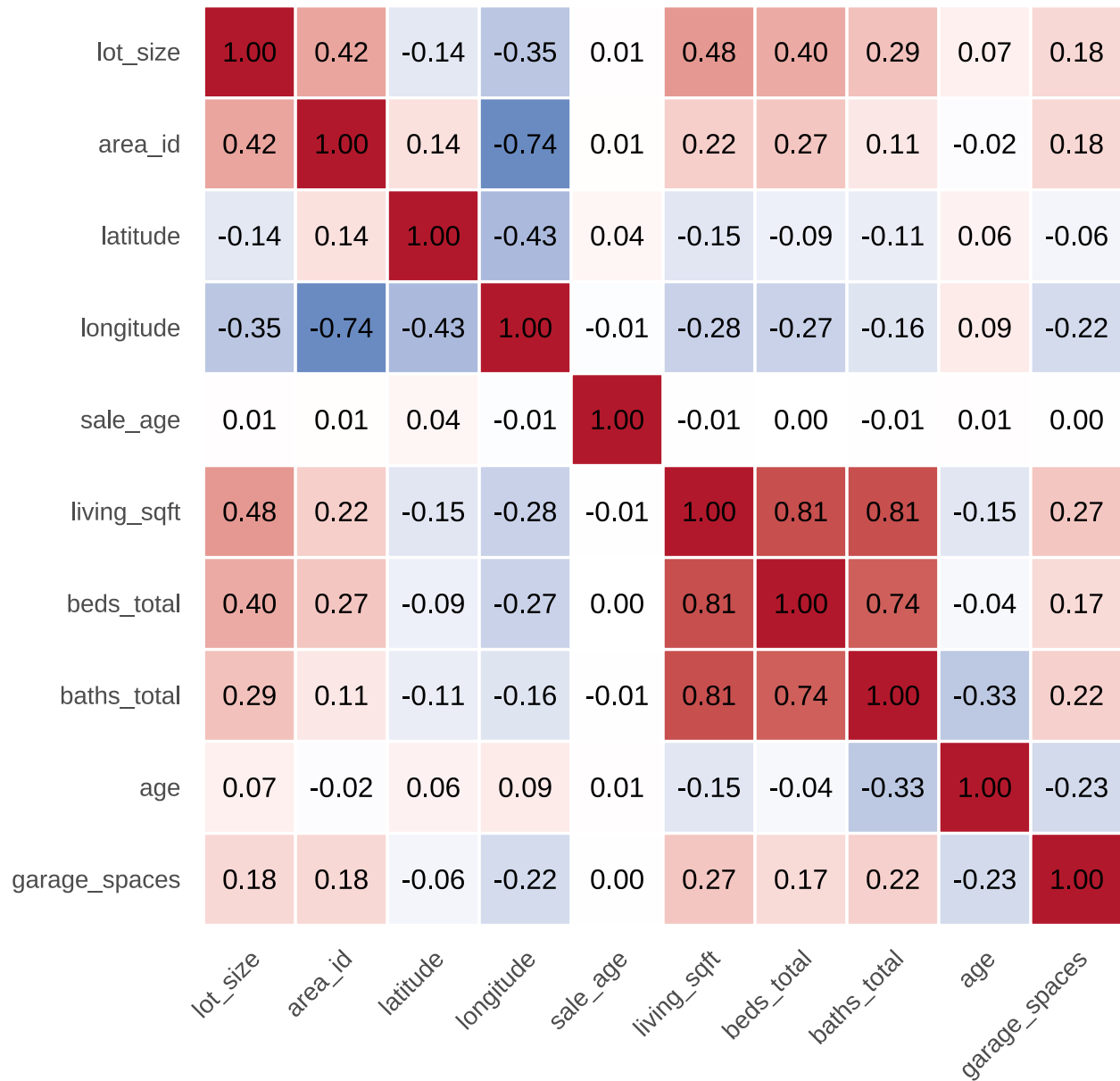


latitude



Correlation Matrix

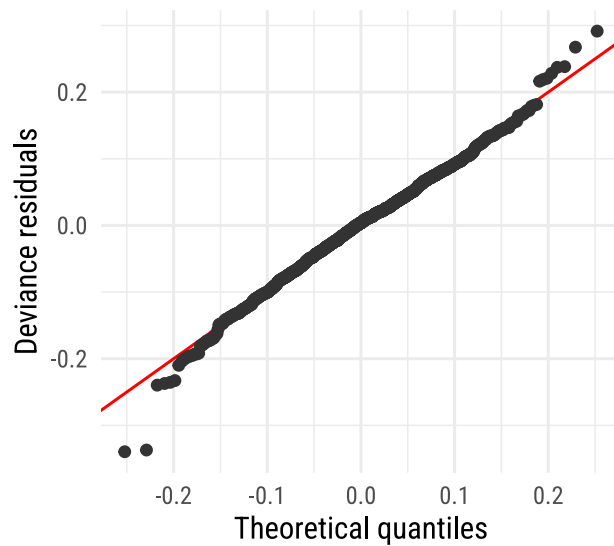
Correlation Matrix



Diagnostics

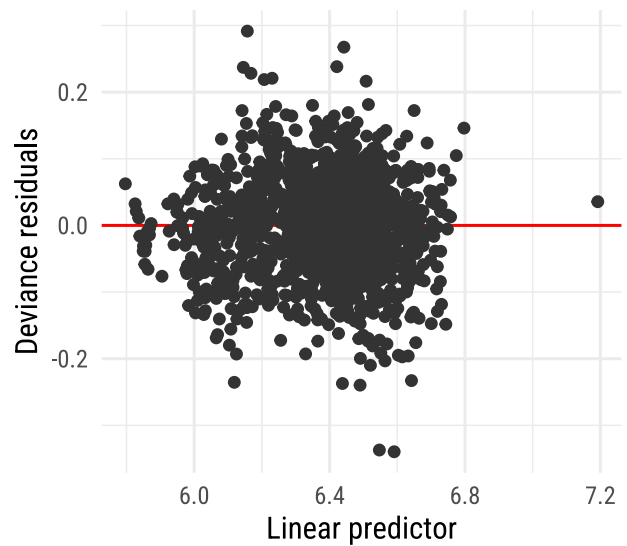
QQ plot of residuals

Method: uniform

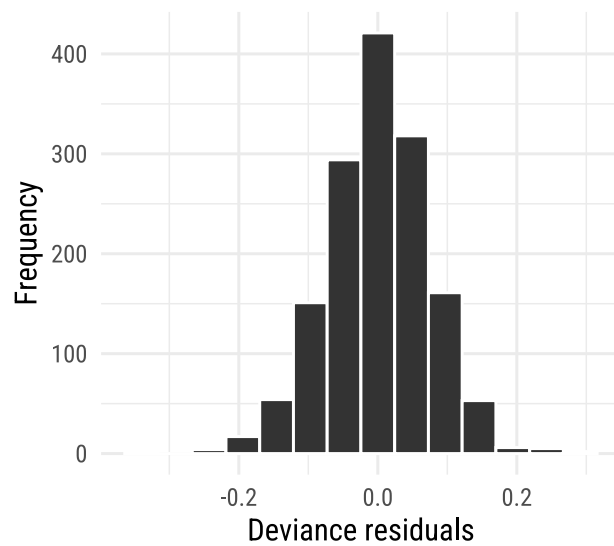


Residuals vs linear predictor

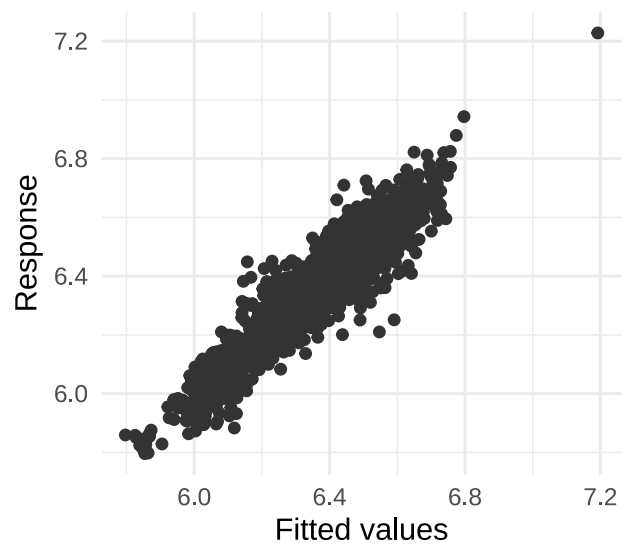
Family: gaussian



Histogram of residuals

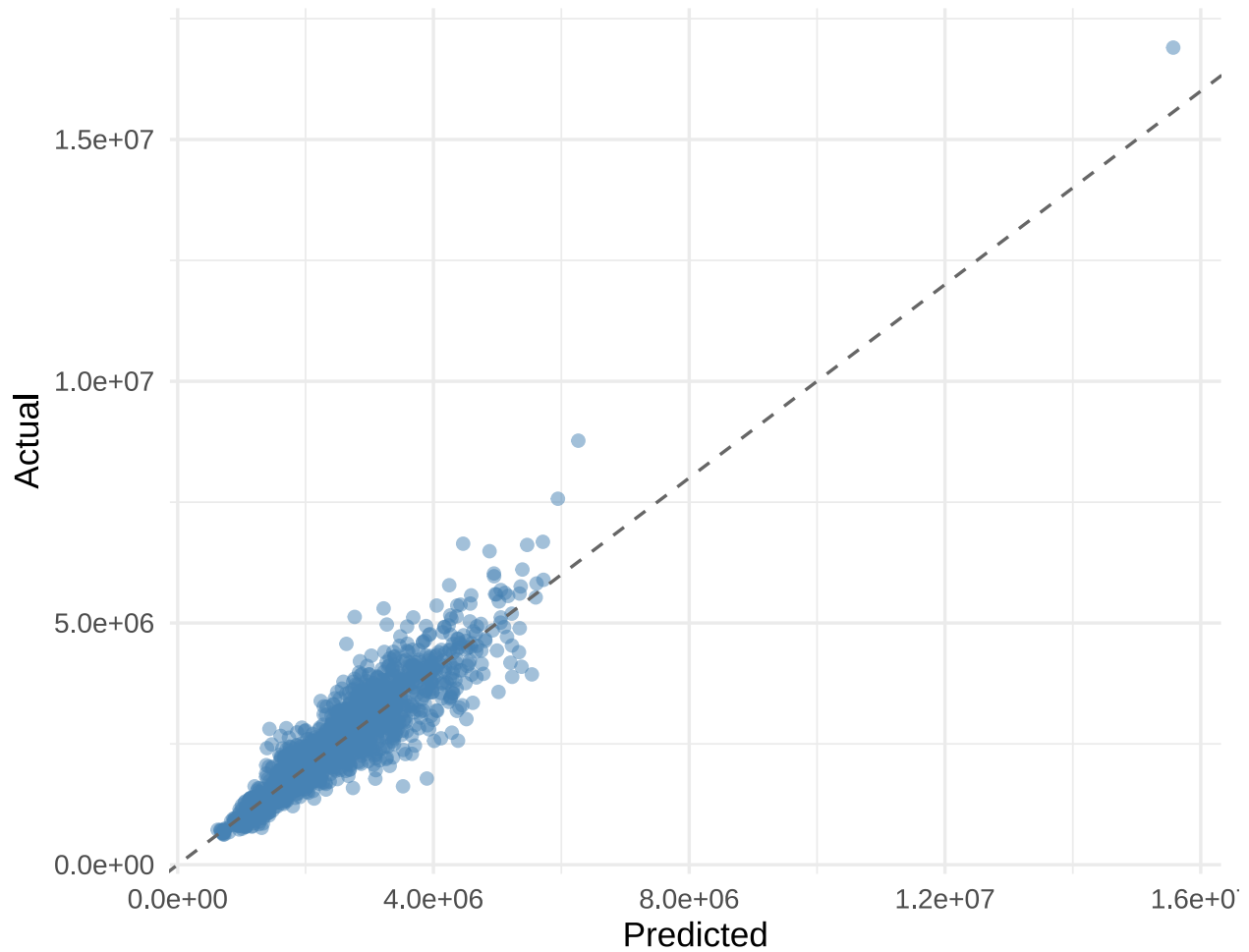


Observed vs fitted values



Actual vs Predicted

Actual vs Predicted (original scale)



ANOVA

Parametric Terms

##

##

##

Term	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-31.2954	17.8843	-1.7499	0.080350
lot_size	0.0000	0.0000	3.7462	0.000187
area_id	-0.0005	0.0009	-0.5774	0.563766
latitude	-0.1711	0.3440	-0.4975	0.618932
longitude	-0.3652	0.1899	-1.9235	0.054616

##

##

Smooth Terms

##

##

##

Term	edf	Ref.df	F	p-value
------	-----	--------	---	---------

```
## |:-----|-----:|-----:|-----:|:-----|
## |s(sale_age)          | 4.5246|      5| 35.2169|< 2e-16 |
## |s(living_sqft)       | 4.8951|      9| 46.2018|< 2e-16 |
## |s(beds_total)        | 2.0668|      7|  2.7456|4.95e-07 |
## |s(baths_total)       | 0.0001|      3|  0.0000|0.918371 |
## |s(age)               | 7.2971|      9| 59.7832|< 2e-16 |
## |s(garage_spaces)     | 2.4215|      4|  3.5927|0.000515 |
## |ti(living_sqft,sale_age) | 4.7602|     16|  3.4256|< 2e-16 |
## |ti(beds_total,living_sqft) | 5.2900|     16|  2.7023|< 2e-16 |
## |ti(baths_total,living_sqft) | 3.0308|     16|  4.1532|< 2e-16 |
```

Concurvity

Overall

Term	worst	observed	estimate
para	1.0000	1.0000	1.0000
s(sale_age)	0.5998	0.3636	0.4134
s(living_sqft)	1.0000	0.9988	0.9807
s(beds_total)	1.0000	0.9970	0.9893
s(baths_total)	0.9993	0.9967	0.9838
s(age)	0.5611	0.4807	0.4022
s(garage_spaces)	0.9451	0.5412	0.3041
ti(living_sqft,sale_age)	0.9977	0.2362	0.0888
ti(beds_total,living_sqft)	1.0000	0.9630	0.9108
ti(baths_total,living_sqft)	1.0000	0.9532	0.8509

Pairwise (worst case)

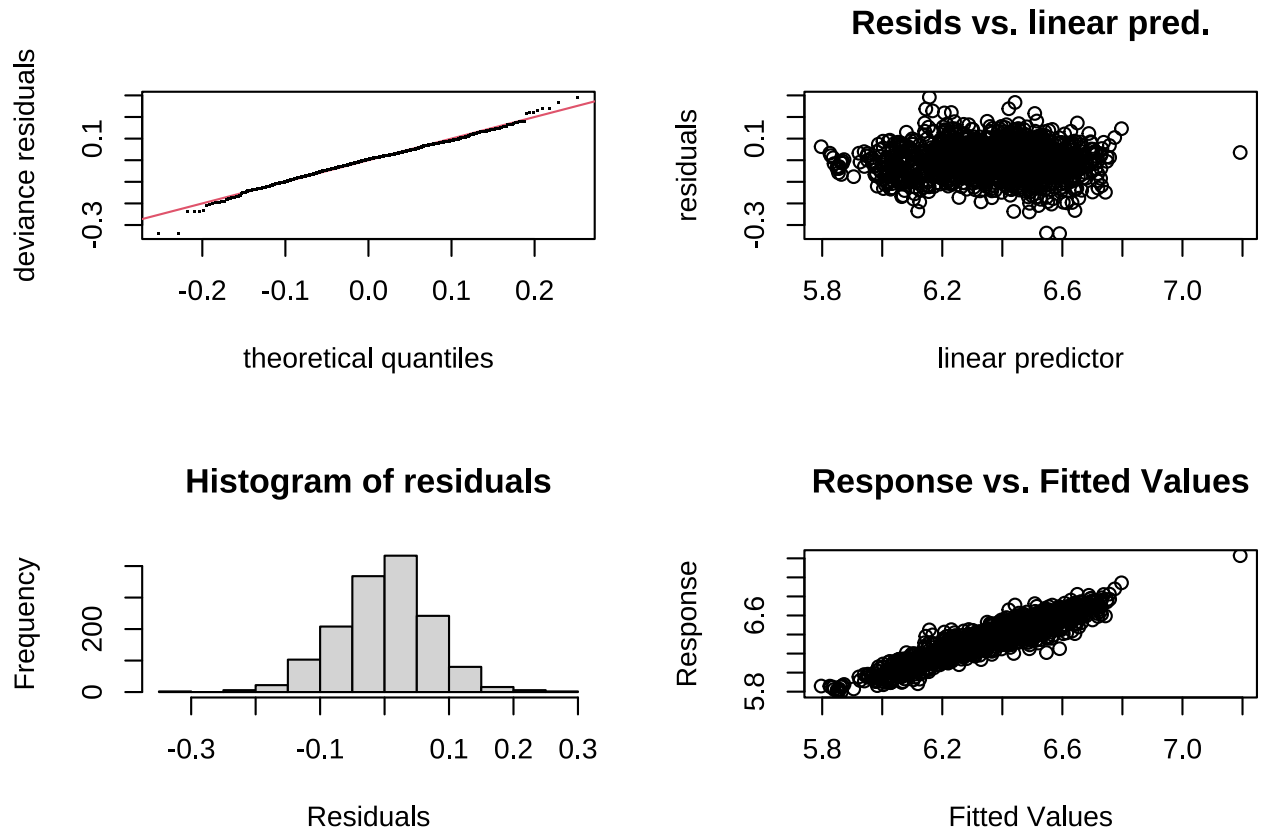
Term	para	sale_age	living_sqft	beds_total	baths_total	garage_spaces	ti(living_sqft,sale_age)	ti(beds_total,living_sqft)	ti(baths_total,living_sqft)
para	1.00	0.00	0.00	0.00	0.00	0.00	0.01	0.70	0.59
sale_age	0.00	1.00	0.01	0.01	0.01	0.02	0.02	0.02	0.02
living_sqft	0.00	0.01	1.00	0.73	0.69	0.20	0.96	1.00	1.00
beds_total	0.00	0.01	0.73	1.00	0.58	0.28	0.94	1.00	0.99
baths_total	0.00	0.01	0.69	0.58	1.00	0.28	0.23	0.50	0.99
age	0.00	0.02	0.20	0.28	0.28	1.00	0.07	0.21	0.30
garage_spaces	0.00	0.01	0.33	0.10	0.06	0.17	0.28	0.71	0.46
ti(living_sqft,sale_age)	0.01	0.02	0.96	0.94	0.23	0.07	1.00	0.98	0.99
ti(beds_total,living_sqft)	0.01	0.02	1.00	1.00	0.50	0.21	0.98	1.00	1.00
ti(baths_total,living_sqft)	0.01	0.02	1.00	0.99	0.99	0.30	0.99	1.00	1.00

Earth vs GAM Sign Consistency

variable	earth_direction	gam_direction	consistent
sale_age	increasing	decreasing	FALSE
living_sqft	increasing	increasing	TRUE
beds_total	mixed	mixed	TRUE
baths_total	increasing	decreasing	FALSE
age	increasing	mixed	TRUE
latitude	increasing	mixed	TRUE

variable	earth_direction	gam_direction	consistent
longitude	mixed	mixed	TRUE
garage_spaces	increasing	mixed	TRUE

Basis Dimension Check



```
##
## Method: REML   Optimizer: outer newton
## full convergence after 14 iterations.
## Gradient range [-0.0008413428,0.002818662]
## (score -1169.675 & scale 0.005499563).
## Hessian positive definite, eigenvalue range [1.684038e-05,528.9815].
## Model rank = 90 / 90
##
## Basis dimension (k) checking results. Low p-value (k-index<1) may
## indicate that k is too low, especially if edf is close to k'.
##
##           k'      edf k-index p-value
## s(sale_age)    5.00e+00 4.52e+00   0.95  0.030 *
## s(living_sqft)  9.00e+00 4.90e+00   0.94  0.010 **
## s(beds_total)   7.00e+00 2.07e+00   0.86 <2e-16 ***
## s(baths_total)  3.00e+00 8.51e-05   0.87 <2e-16 ***
## s(age)         9.00e+00 7.30e+00   0.87 <2e-16 ***
## s(garage_spaces) 4.00e+00 2.42e+00   0.93  0.005 **
## ti(living_sqft,sale_age) 1.60e+01 4.76e+00   0.99  0.385
## ti(beds_total,living_sqft) 1.60e+01 5.29e+00   1.00  0.570
## ti(baths_total,living_sqft) 1.60e+01 3.03e+00   0.97  0.130
```



```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Mgcv Output

```
== Timing: 1.69 seconds ==
```

```
== Formula ==
```

```
sale_price ~ s(sale_age, bs = "cr", k = 6) + s(living_sqft, bs = "tp") +
  s(beds_total, bs = "tp", k = 8) + s(baths_total, bs = "cr",
    k = 4) + lot_size + area_id + s(age, bs = "tp") + latitude +
  longitude + s(garage_spaces, bs = "tp", k = 5) + ti(living_sqft,
    sale_age, bs = c("tp", "tp")) + ti(beds_total, living_sqft,
    bs = c("tp", "tp")) + ti(baths_total, living_sqft, bs = c("tp",
    "tp"))
```

```
<environment: 0x129c2be60>
```

```
== Summary ==
```

```
Family: gaussian
```

```
Link function: identity
```

```
Formula:
```

```
sale_price ~ s(sale_age, bs = "cr", k = 6) + s(living_sqft, bs = "tp") +
  s(beds_total, bs = "tp", k = 8) + s(baths_total, bs = "cr",
    k = 4) + lot_size + area_id + s(age, bs = "tp") + latitude +
  longitude + s(garage_spaces, bs = "tp", k = 5) + ti(living_sqft,
    sale_age, bs = c("tp", "tp")) + ti(beds_total, living_sqft,
    bs = c("tp", "tp")) + ti(baths_total, living_sqft, bs = c("tp",
    "tp"))
```

```
Parametric coefficients:
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-3.130e+01	1.788e+01	-1.750	0.080350 .
lot_size	1.736e-06	4.635e-07	3.746	0.000187 ***
area_id	-5.217e-04	9.036e-04	-0.577	0.563766
latitude	-1.711e-01	3.440e-01	-0.497	0.618932
longitude	-3.652e-01	1.899e-01	-1.923	0.054616 .

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Approximate significance of smooth terms:
```

	edf	Ref.df	F	p-value
s(sale_age)	4.525e+00	5	35.217	< 2e-16 ***
s(living_sqft)	4.895e+00	9	46.202	< 2e-16 ***
s(beds_total)	2.067e+00	7	2.746	4.95e-07 ***
s(baths_total)	8.508e-05	3	0.000	0.918371
s(age)	7.297e+00	9	59.783	< 2e-16 ***
s(garage_spaces)	2.421e+00	4	3.593	0.000515 ***
ti(living_sqft,sale_age)	4.760e+00	16	3.426	< 2e-16 ***
ti(beds_total,living_sqft)	5.290e+00	16	2.702	< 2e-16 ***
ti(baths_total,living_sqft)	3.031e+00	16	4.153	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

R-sq.(adj) = 0.859 Deviance explained = 86.3%
-REML = -1169.7 Scale est. = 0.0054996 n = 1488

== Family ==

Family: gaussian
Link function: identity

Report generated by mgcvUI on 2026-03-28 01:36:56.687928